

Task A

Crucial Notice before you read and mark my work

Mac-specific session setup. macOS BSD tools (cut, grep, tr) error out on non-UTF-8 bytes in the property descriptions, producing cut: stdin: Illegal byte sequence and silently truncated output.

The fix is export LC_ALL=C, which switches the shell to byte-mode.

Note: this only lasts for the current terminal session — closing the terminal resets it. You'll see I add `export...` as step 0 at the beginning of each questions code block.

Q1 — sold_time range of the records

Approach

`sold_time` has the form "`D/M/YYYY H:MM`" but is **not** zero-padded format (e.g. `8/11/2021 16:33` sits beside `27/03/2021 22:17`), which leads to wrong `sort`. So convert each value to a sortable form `YYYY-MM-DD HH:MM` before sorting.

Code

```
cd ~/Documents/5145/Assignment4/TaskA_shell

# Switch the whole session to byte-mode (fixes the Illegal byte sequence)
export LC_ALL=C

# Confirm — should now print "C"
echo $LC_ALL

tail -n +2 TaskA_property_victoria.csv | cut -d, -f4 | tr -d '\r' | grep -E
'^[0-9]+/[0-9]+/[0-9]+ [0-9]+:[0-9]+$' > q1_sold_times.txt

wc -l q1_sold_times.txt    # 12699

# Investigate years present in the data
awk -F'[ /:]' '{ print $3 }' q1_sold_times.txt | sort | uniq -c
# Output: 1 record from 2020, 127698 from 2021
# Because there is only one record on 2020 (account for 1/127699 =
# 0.0008%). And the assignment specification which states the dataset covers
# 2021.
# Treat as a data-entry anomaly and exclude from the range.

awk -F'[ /:]' '$3 == "2020"' q1_sold_times.txt
# Output: 01/01/2020 00:23 — note the suspiciously round time (00:23 on
# 1 January). Likely a year typo (2020 instead of 2021).

# Recompute the range after excluding the 2020 anomaly
```

```

awk -F'[ /:]' '$3 != "2020"' q1_sold_times.txt | awk -F'[ /:]' '{ printf
"%04d-%02d-%02d %02d:%02d\t%s\n", $3, $2, $1, $4, $5, $0 }' | sort >
q1_sorted_clean.txt

echo "Earliest (after excluding 2020 anomaly): $(head -n 1
q1_sorted_clean.txt | cut -f2)"
echo "Latest (after excluding 2020 anomaly): $(tail -n 1
q1_sorted_clean.txt | cut -f2)"

awk -F'[ /:]' '{ printf "%04d-%02d-%02d %02d:%02d\t%s\n", $3, $2, $1, $4,
$5, $0 }' q1_sold_times.txt | sort > q1_sorted.txt

echo "Earliest: $(head -n 1 q1_sorted.txt | cut -f2)"
echo "Latest: $(tail -n 1 q1_sorted.txt | cut -f2)"

```

Screenshot

```

(base) ez_us@MacBook-Pro-5 TaskA_shell % echo "Earliest: $(head -n 1 q1_sorted.txt | cut -f2)"
Earliest: 01/01/2020 00:23
(base) ez_us@MacBook-Pro-5 TaskA_shell % echo "Latest: $(tail -n 1 q1_sorted.txt | cut -f2)"
Latest: 31/12/2021 23:58

(base) ez_us@MacBook-Pro-5 TaskA_shell % echo "Earliest (after excluding 2020 anomaly): $(head -n 1 q1_sorted_clean.txt | cut -f2)"
Earliest (after excluding 2020 anomaly): 1/01/2021 0:19
(base) ez_us@MacBook-Pro-5 TaskA_shell % echo "Latest (after excluding 2020 anomaly): $(tail -n 1 q1_sorted_clean.txt | cut -f2)"
Latest (after excluding 2020 anomaly): 31/12/2021 23:58

```

Answer

Raw range: 01/01/2020 00:23 - 31/12/2021 23:58. Anomaly identified: the single record 01/01/2020 00:23 is the only 2020 entry in the dataset (1 row out of 127,699 valid dates = 0.0008%). The assignment specification states the dataset contains transactions from 2021. The round time (00:23 on 1 January) and singleton year strongly suggest a year-typo data-entry error. Range after excluding the anomaly: 1 January 2021 at 00:19 - 31 December 2021 at 23:58.

Explanation

- Isolate `sold_time` column.
 - `cut -d, -f4`, cut the 4th column according to assignment description.
 - `tr -d 'r'`, removes 'return' character.
 - `grep ...` filter rows with valid time format.
 - output it as a new text file: `q1_sold_times.txt`, which include 127699 data.
- Sort `sold_times`.
 - Standardise time format. current date would look like "M/D/YYYY H:S" or "MM/DD/YYYY HH:SS", Which determined by whether have "0" or not on date, and 24-hour time standard. It lead to disorder. Parse date form first, then sort.
 - `awk -F'[ /:]'` splits a string like "27/03/2021 22:17" on any of space, slash, or colon — so we get the five fields 27, 03, 2021, 22, 17.

- Then `printf"%04d-%02d-%02d %02d:%02d\t%s\n", $3(year), $2(month), $1(day), $4(hour), $5(minute), $0(Placeholder: No second here)` rearranges them year-first and prints both the sortable form and the original.
- `sort` orders the lines alphabetically — but because our key is zero-padded and year-first, alphabetical order IS chronological order.
- Strip off the first and last line to get answers
 - `echo` can be regarded as `print` or `cat`(R)

Q2 - Preprocess the ID(f1) and sold_time(f4) column.

Q2a - Count lines aren't 6-digit long.

Approach

regex `^[0-9]{6}$` would be valid ID

Code

```
# Step 0: disable strict UTF-8 (BSD cut fails otherwise)
export LC_ALL=C

# Step 1: extract ID column, skip header
tail -n +2 TaskA_property_victoria.csv | cut -d, -f1 > q2a_ids.txt

# Step 2: count rows whose ID isn't a 6-digit long number
# grep -v inverts the match; -c means 'count'
grep -vcE '^[0-9]{6}$' q2a_ids.txt

grep -vE '^[0-9]{6}$' q2a_ids.txt | sort | uniq -c
```

Screenshot

```
((base) ez_us@MacBook-Pro-5 TaskA_shell % grep -vcE '^[0-9]{6}$' q2a_ids.txt
9
((base) ez_us@MacBook-Pro-5 TaskA_shell % grep -vE '^[0-9]{6}$' q2a_ids.txt | sort | uniq -c
 1 122
 1 15049
 1 17176
 1 2396
 5 NA
```

Answer

9 rows have an **ID** that is not a 6-digit number: 5 rows where the ID is the literal string **NA**, and 4 rows whose IDs are short numbers (**122**, **2396**, **15049**, **17176**).

Explanation

- `grep -v` keeps lines that **do not** match the pattern. `-c` return to the number of count.
- `^[0-9]{6}` means 6 digits. `$` limited the output **exactly** equal to 6.
- `sort | uniq -c` arrange **unique count** value from small to large.

Q2b — Drop bad-ID rows and strip the time from sold_time

Approach

Two transformations:

1. Keep the header plus rows valid ID, which matches `^[0-9]{6}$`.
2. In each kept row, remove the `HH:MM` substring from column 4.

Code

```
# Step 0: disable strict UTF-8 (BSD cut fails otherwise)
export LC_ALL=C

# Step 1: extract a valid ID. Keep header (NR==1) and rows with 6-digit
long form.
awk -F, 'NR==1 || $1 ~ /^[0-9]{6}$/' TaskA_property_victoria.csv >
q2b_valid_ids.csv

wc -l q2b_valid_ids.csv # check

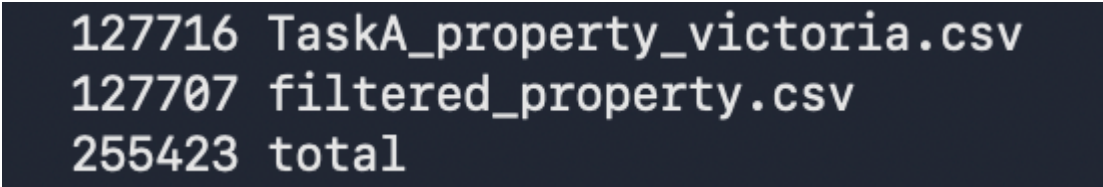
# output: 1 header + 127706 data rows = 127707

awk -F, 'BEGIN { OFS = "," } NR == 1 { print; next } { sub(/ [0-9]+:[0-9]+/, "", $4); print }' q2b_valid_ids.csv > filtered_property.csv

wc -l filtered_property.csv # check
# output: 1 header + 127706 data rows = 127707. No miss.

# Verify:
# original 127716 - 9 bad rows = 127707 lines (header + 127706 data rows)
wc -l TaskA_property_victoria.csv filtered_property.csv
```

Screenshot



```
127716 TaskA_property_victoria.csv
127707 filtered_property.csv
255423 total
```

Explanation

- `awk -F, 'NR==1 || $1 ~ /^[0-9]{6}$/'` is the entire pattern.
 - `NR==1` is true on the header row.
 - `$1 ~ /^[0-9]{6}$` is true on rows where the ID is exactly 6 digits.
 - `||` means "or".
- `sub(/ [0-9]+:[0-9]+/, "", $4)` finds the first occurrence of "H:MM" or "HH:MM" inside column 4 and replaces it with empty. Because we assign to a field, `awk` rebuilds the line `OFS = ","`: Since the input and output separators are both comma, every other field comes back byte-identical, including rows `description` contained unescaped commas.
- `127716 - 9 = 127707` — counts match Q2a.

Q2c — Display first 5 lines of ID and `sold_time`

Code

```
export LC_ALL=C

cut -d, -f1,4 filtered_property.csv | head -n 6
```

Screenshot

```
((base) ez_us@MacBook-Pro-5 TaskA_shell % cut -d, -f1,4 filtered_property.csv | head -n 6
ID,sold_time
294290,27/03/2021
169586,18/02/2021
237723,29/04/2021
116018,8/11/2021
210091,27/10/2021
```

Explanation

- `cut -d, -f1,4` picks columns 1 and 4 (`ID` and `sold_time`); they're safe with `cut` because no commas appear inside the first four columns.
- `head -n 6` 1 header + the first 5 data rows

Q3 — First and last mention of "Mount Dandenong" in `address`

Approach

Important note for marker: This question uses `filtered_property.csv`, the cleaned dataset produced in Q2b (bad-ID rows dropped, time stripped from `sold_time`). The filtering performed in Q2b is the prerequisite for all subsequent questions in Task A.

Use `awk` to keep just the rows whose column 7(`address`) contains the literal `Mount Dandenong`. Then sort date.

Code

```
# Step 0: Switch the whole session to byte-mode (fixes the Illegal byte
sequence)
export LC_ALL=C

# Step 1: keep only rows whose address (col 7) contains "Mount Dandenong"
awk -F, '$7 ~ /Mount Dandenong/' filtered_property.csv > q3_matches.csv

wc -l q3_matches.csv                # how many transactions matched - Ans:
44

# Step 2: extract sold_time (col 4), drop any stray CRs
cut -d, -f4 q3_matches.csv | tr -d '\r' > q3_dates.txt

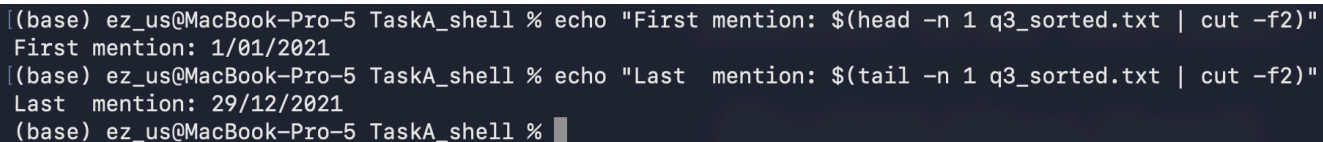
head -n 3 q3_dates.txt             # sanity check: these should be DD/MM/YYYY

# Step 3: convert "D/M/YYYY" to "YYYY-MM-DD<TAB>original", sort
awk -F/ '{ printf "%04d-%02d-%02d\t%s\n", $3, $2, $1, $0 }' q3_dates.txt |
sort > q3_sorted.txt

head -n 2 q3_sorted.txt

# Step 4: read off first and last
echo "First mention: $(head -n 1 q3_sorted.txt | cut -f2)"
echo "Last  mention: $(tail -n 1 q3_sorted.txt | cut -f2)"
```

Screenshot



```
((base) ez_us@MacBook-Pro-5 TaskA_shell % echo "First mention: $(head -n 1 q3_sorted.txt | cut -f2)"
First mention: 1/01/2021
((base) ez_us@MacBook-Pro-5 TaskA_shell % echo "Last  mention: $(tail -n 1 q3_sorted.txt | cut -f2)"
Last  mention: 29/12/2021
(base) ez_us@MacBook-Pro-5 TaskA_shell %
```

Answer

- **First** (earliest) mention of "Mount Dandenong" in `address`: **1 January 2021**
- **Last** (latest) mention: **29 December 2021**

Explanation

- `awk -F, '$7 ~ /Mount Dandenong/'` column 7 contains the `Mount Dandenong`. Awk regexes are case-sensitive, so this matches the spec's requirement.
- `cut -d, -f4` extracts the `sold_time` column (now a date with no time, after Q2b). `tr -d '\r'` removes any `return`.

- Sort year-month-day.
- `echo` earliest an last line.

Q4 — First/last sold_time: odd month, Townhouse, area > 300

Important note for marker: This question uses filtered_property.csv, the cleaned dataset produced in Q2b (bad-ID rows dropped, time stripped from sold_time). The filtering performed in Q2b is the prerequisite for all subsequent questions in Task A.

Approach

Filter `odd month, Townhouse, area > 300`, remove `NA`. Then Sort and Print.

Code

```
# Step 0: Switch the whole session to byte-mode (fixes the Illegal byte
sequence)
export LC_ALL=C

# Step 1: keep only Townhouse rows
awk -F, '$12 == "Townhouse"' filtered_property.csv > q4_step1_townhouse.csv
wc -l q4_step1_townhouse.csv # 10471

# Step 2: among Townhouses, keep rows where area is a plain number AND >
300.
# Pattern 1 of $11 matches a non-negative decimal (no NA, no "Xha", no "qq");
# Pattern 2: numeric value of $11 (forced by + 0) is greater than 300
awk -F, '$11 ~ /^[0-9]+(\.[0-9]+)?$/ && ($11 + 0) > 300'
q4_step1_townhouse.csv > q4_step2_area.csv
wc -l q4_step2_area.csv # 814

# Step 3: keep only rows whose sold_time month is odd (1, 3, 5, 7, 9, 11)
# split($4, d, "/") gives d[1]=DD, d[2]=MM, d[3]=YYYY
# (d[2] + 0) % 2 == 1 is true only for odd months
awk -F, '{split($4, d, "/"); if((d[2] + 0) % 2 == 1) print}'
q4_step2_area.csv > q4_step3_oddmonth.csv
wc -l q4_step3_oddmonth.csv # 398

# Step 4: extract sold_time, build sortable key, sort
cut -d, -f4 q4_step3_oddmonth.csv | tr -d '\r' | awk -F/ '{ printf "%04d-
%02d-%02d\t%s\n", $3, $2, $1, $0 }' | sort > q4_sorted.txt

# Step 5: report first and last
echo "First sold_time: $(head -n 1 q4_sorted.txt | cut -f2)"
echo "Last sold_time: $(tail -n 1 q4_sorted.txt | cut -f2)"
```

Screenshot


```
(base) ez_us@MacBook-Pro-5 TaskA_shell % echo "First sold_time: $(head -n 1 q4_sorted.txt | cut -f2)"
First sold_time: 4/01/2021
(base) ez_us@MacBook-Pro-5 TaskA_shell % echo "Last sold_time: $(tail -n 1 q4_sorted.txt | cut -f2)"
Last sold_time: 30/11/2021
```

Answer

- **First** matching **sold_time: 4 January 2021**
- **Last** matching **sold_time: 30 November 2021**

Explanation

- Filter (1) Townhouse (2) area > 300 (3) odd month.
- Process **NA**, and other wrong valua filter by regex `^[0-9]+(\.[0-9]+)?$. $11 + 0` forces numeric context so the comparison is numerical (so `"800" > 300` is true and `"50" > 300` is false; without `+ 0`, awk would compare as strings and get `"50" > "300"` as true, which is wrong).
- `split($4, d, "/")` splits `"4/01/2021"` into `d[1]=4`, `d[2]=01`, `d[3]=2021`. `(d[2] + 0) % 2 == 1` find odd month. `+ 0` force ensure data format is number. i.e. `"01"` to be treated as the number 1, not as a string.

Q5 — How many unique property types?

Importand note for marker: This question uses `filtered_property.csv`, the cleaned dataset produced in Q2b (bad-ID rows dropped, time stripped from `sold_time`). The filtering performed in Q2b is the prerequisite for all subsequent questions in Task A.

Approach

The answer is the number of distinct values in `property_type`, but inspection shows the column is **highly contaminated** with `area` numbers, an address fragment, and an `NA`. So report the raw count first *and* a cleaned count, with the wrangling spelled out.

Code

```
# Step 0: Switch the whole session to byte-mode (fixes the Illegal byte
sequence)
export LC_ALL=C

# Step 1: extract property_type column (col 12), skip header
tail -n +2 filtered_property.csv | cut -d, -f12 > q5_types.txt
wc -l q5_types.txt    # 127706

# Step 2: raw distinct count
sort -u q5_types.txt | wc -l    # 390

# Step 3: see the counts of various type – common types first, then the
tail
sort q5_types.txt | uniq -c | sort -rn > q5_unique_v1.txt
```



```

head -n 15 q5_unique_v1.txt # top 15 most common
tail -n 15 q5_unique_v1.txt # bottom 15. You can see them is meaningless. So
filter out them next. Data wranling needed!

# Step 4: filter out garbage data(tail-like rows). Each grep -vE drops one
category:
# - "^NA$"
# - "^[0-9]+(\.[0-9]+)?(ha)?$" - pure numbers and 'Xha'
# - "^[0-9]+ .* VIC " - address-like (e.g. "20 ... VIC 3984")
sort -u q5_types.txt | grep -vE '^NA$' | grep -vE '^[0-9]+(\.[0-9]+)?(ha)?
$' | grep -vE '^[0-9]+ .* VIC ' > q5_unique_v2.txt

wc -l q5_unique_v2.txt # 34
cat q5_unique_v2.txt

# Step 5: case-collapsed count ("Vacant Land" and "Vacant land" merge)
tr 'A-Z' 'a-z' < q5_unique_v2.txt | sort -u | wc -l

```

Screenshot

- Step 2

```

[(base) ez_us@MacBook-Pro-5 TaskA_shell % sort -u q5_types.txt | wc -l
390

```

- Gabage Data format

- Address

```

q5_unique_v1.txt:329: 1 20 Paul Street Grantville VIC 3984

```

- o Number and NA type

```
(base) ez_us@MacBook-Pro-5 TaskA_shell % head -n 100 q5_unique_v1.txt
80923 House
20972 Apartment / Unit / Flat
10471 Townhouse
10236 Vacant land
1682 Rural
1018 Acreage / Semi-Rural
378 Villa
317 New House & Land
238 NA
191 Farm
155 Block of Units
150 New land
134 New Apartments / Off the Plan
85 Specialist Farm
79 Studio
45 Rural Lifestyle
35 Development Site
33 Carspace
22 Semi-Detached
21 Livestock
16 Terrace
13 Duplex
12 4000
10 Retirement
10 Mixed Farming
9 8.09ha
9 1
5 392
5 343
5 2000
4 Farmlet
4 Dairy Farming
4 862
4 796
4 2.02ha
4 2
4 1022
3 930
3 667
3 576
3 509
3 4
```

- Step 3

```
((base) ez_us@MacBook-Pro-5 TaskA_shell % sort q5_types.txt | uniq -c | sort -rn > q5_unique_v1.txt
((base) ez_us@MacBook-Pro-5 TaskA_shell % head -n 15 q5_unique_v1.txt
80923 House
20972 Apartment / Unit / Flat
10471 Townhouse
10236 Vacant land
1682 Rural
1018 Acreage / Semi-Rural
378 Villa
317 New House & Land
238 NA
191 Farm
155 Block of Units
150 New land
134 New Apartments / Off the Plan
85 Specialist Farm
79 Studio
((base) ez_us@MacBook-Pro-5 TaskA_shell % tail -n 15 q5_unique_v1.txt
1 1033
1 103.5
1 1021
1 1016
1 1013
1 1011.83
1 1011
1 1.76ha
1 1.62ha
1 1.4ha
1 1.44ha
1 1.42ha
1 1.29ha
```

- Check result if they are purely unique or exist

```
(base) ez_us@MacBook-Pro-5 TaskA_shell % cat q5_unique_v2.txt
Acreage / Semi-Rural
Apartment / House / Dairy Farming
Apartment / Unit / Flat
Block of Units
Carspace
Cropping
Dairy Farming
Development Site
Duplex
Duplex
Farm
Farmlet
House
Livestock
Mixed Farming
New Apartments / Off the Plan
New Home Designs
New House & Land
New land
Penthouse
Retirement
Rural
Rural Lifestyle
Semi-Detached
Sky Villa
Smart Pod
Specialist Farm
Studio
Terrace
Townhouse
Vacant Land
Vacant land
Villa
Viticulture
(base) ez_us@MacBook-Pro-5 TaskA_shell %
```

They are
same thing

```
(base) ez_us@MacBook-Pro-5 TaskA_shell % tr 'A-Z' 'a-z' < q5_unique_v2.txt | sort -u | wc -l
33
```

Answer

- **Raw distinct count:** 390 distinct strings in `property_type`. = Step 2
- **unique_v1 property types after removing garbage data(step 3): 34.**
- **After standarding case variants (Vacant Land ↔ Vacant land): 33.**

I report **34** (or **33**) as the meaningful answer, because the other 356 raw values are not property types — 354 are `area` numbers misplaced into the column, 1 is `NA`, and 1 is an address string.

Explanation

- `-u` find the unique value, Extract to single test file. `wc -l` counts lines of the new file.

- `sort | uniq -c | sort -rn` chain — sort alphabetically so duplicates are adjacent, `uniq -c` counts each group, `sort -rn` re-sorts by count (numeric, big to small) so the most common types come first. Check the head for valid property names and the tail for meaningless data(garbage).
- Filtering: Using `grep -vE` to remove the three patterns target the three garbage shapes spotted in step 3.
- Collapse: `tr 'A-Z' 'a-z'` converts(transfer) uppercase to lowercase; then `sort -u | wc -l` recounts distinct values. `Vacant Land` (1 row) and `Vacant land` (10,236 rows) merge, dropping the count from 34 to 33.

Q6 — Description column investigations

Important note for marker: This question uses `filtered_property.csv`, the cleaned dataset produced in Q2b (bad-ID rows dropped, time stripped from `sold_time`). The filtering performed in Q2b is the prerequisite for all subsequent questions in Task A.

Ground setup: extract description text for searching

So extract the 14th - `description` - column first

```
# Step 0: Switch the whole session to byte-mode (fixes the Illegal byte
sequence)
export LC_ALL=C

# Build a description-only file used by Q6a and Q6b, and transfer clean to
lowercase format(It mentioned ignore cases in question)
tail -n +2 filtered_property.csv | cut -d, -f14 | tr 'A-Z' 'a-z' >
q6_descriptions.txt

wc -l q6_descriptions.txt # 127706
```

Q6a — Premium/luxury properties with outdoor entertaining areas

Code

```
# Step 1: keep descriptions premium OR luxury
grep -E '(premium|luxury)' q6_descriptions.txt > q6a_prem_lux.txt
wc -l q6a_prem_lux.txt

# Step 2: among those, keep ones mentioning "outdoor entertaining area"
grep 'outdoor entertaining area' q6a_prem_lux.txt > q6a_prem_lux_out.txt
wc -l q6a_prem_lux_out.txt
```


Screenshot

```
(base) ez_us@MacBook-Pro-5 TaskA_shell % wc -l q6_descriptions.txt
127706 q6_descriptions.txt
(base) ez_us@MacBook-Pro-5 TaskA_shell % grep -E '(premium|luxury)' q6_descriptions.txt > q6a_prem_lux.txt
(base) ez_us@MacBook-Pro-5 TaskA_shell % wc -l q6a_prem_lux.txt
13642 q6a_prem_lux.txt
(base) ez_us@MacBook-Pro-5 TaskA_shell % grep 'outdoor entertaining area' q6a_prem_lux.txt > q6a_prem_lux_out.txt
(base) ez_us@MacBook-Pro-5 TaskA_shell % wc -l q6a_prem_lux_out.txt
224 q6a_prem_lux_out.txt
(base) ez_us@MacBook-Pro-5 TaskA_shell %
```

Answer

224 transaction records describe premium-or-luxury properties with an outdoor entertaining area.

Explanation

- Built `q6_descriptions.txt` once and reuse it in both Q6a and Q6b. `tr [A-Z] [a-z]` lowercase to ignore cases,
- **Step 1:** `grep -E '(premium|luxury)'` keeps lines containing either keyword. `-E` enables extended regex so the `|` means "or"
- **Step 2:** filter 'outdoor entertaining area'.

Q6b — Records mentioning a property size in their description

Approach

The spec gives two example forms: `Nm2` and `N sq metres`. I treat those as my **reference unit list**. To extend the list, I:

1. consulted standard Australian property-area units (square metre, hectare, acre) from Google,
2. inspected the description column to confirm these forms are present in the data, and to spot any forms I missed.

Code

```
# Detect "digit + unit", "1249m2" likewise string
grep -oE '[0-9]+[a-z]+[0-9]*' q6_descriptions.txt | sort | uniq -c | sort -rn | grep -iE '(m2|sqm|acres?|hectare|ha)' | head -30

# Detect "digit + space + unit word", "758 sq metres" likewise string
grep -oE '[0-9]+ [a-z]+([a-z]+)?' q6_descriptions.txt | sort | uniq -c | sort -rn | grep -iE '(sq|acres?|hectares?|metres?)' | head -30

# Now I know the forms used: m2, sqm, sq metres, square metres, acres, hectares, ha.
# Build the regex from observation:
grep -cE '[0-9]+m2|[0-9]+ ?sqm|[0-9]+ sq metres?|[0-9]+ square metres?|[0-9]+ acres?|[0-9]+ hectares?|[0-9]+ ?ha([.,;]|$)' q6_descriptions.txt
```

Screenshot

```
((base) ez_us@MacBook-Pro-5 TaskA_shell % grep -oE '[0-9]+[a-z]+[0-9]*' q6_descriptions.txt | sort | uniq -c | sort -rn | grep -iE '(m2|sqm|acres?|hectare|ha)' | head -10
238 650m2
231 1000m2
183 700m2
173 800m2
150 448m2
145 400m2
144 600m2
140 560m2
140 4sqm
136 1000sqm
((base) ez_us@MacBook-Pro-5 TaskA_shell % grep -oE '[0-9]+ [a-z]+([a-z]+)?' q6_descriptions.txt | sort | uniq -c | sort -rn | grep -iE '(sq|acres?|hectares?|metres?)' | head -10
374 4 sqm
268 5 acres
171 5 acres of
134 1 acre
109 2 acres
79 10 acres
78 3 acres
76 4 acre
71 2 acres of
70 6 acres
((base) ez_us@MacBook-Pro-5 TaskA_shell % grep -cE '[0-9]+m2|[0-9]+ ?sqm|[0-9]+ sq metres?|[0-9]+ square metres?|[0-9]+ acres?|[0-9]+ hectares?|[0-9]+ ?ha([.,;]|$)' q6_descriptions.txt
42066
```

Answer

42066 transaction records mention property size information in their description.

Cleanup

if marker run code in this work, you can run the below code to remove intermiat files

To remove all intermediate files but keep **filtered_property.csv**, which generate in q2b(Drop bad-ID rows and strip the time from sold_time):

```
rm q*.txt q*_step*.csv q2b_valid_ids.csv q3_matches.csv # here can replace
with the file(s) of qustion you mark

# Snitary check
ls *.csv
```